Good vs Bad Layout Practices

1. OBJECTIVES

- To compare the effects of layout practices in PCB on the power rail by designing identical circuits with good and bad layout practices.
- To check the effect of capacitor in the output regarding noise in the power rail.

2. COMPONENTS USED/BOM

1. NE555DR
2. LDO – AMS1117
3. Hex Inverter - SN74AHC14DR
4. Resistor – 47ohm x6, 1kohm x4, 10kohm x2.
5. LED – 8 nos
6. Power plug
7. Header pins and shorting flags x4.
8. Capacitor – 1uF x1 , 22uF x5
9. Mini Slide switch – 3 pin x 2

3. PLAN OF RECORD (POR)

The PCB circuit should contain the following

- A “AC to 5VDC” Power plug for powering the circuit.
- LED indication to know if power supply is ON
- A LDO to convert 5V to 3.3V which powers the Hex inverters in 3.3V mode.
- LED indication to know if 3.3V is working
- A Isolation switch to control and measure the effects of output capacitor on the3.3V power rail.
- A Mini slide switch to choose between 3.3V or 5V mode to power up the Hex inverters.
- Header pin to isolate 555 timer section from power supply section
- 555 timer to provide 5Vp-p square wave signal with around 500Hz and 66% Duty cycle for clock to the hex inverters.
- A Mini slide switch to choose clock output for good or bad layout hex inverters.
- Two hex inverter circuits with different layout considerations.
- Three LED’s in the output of three inverter pins to draw some load current.
- 10K resistor to pull up inverter inputs to avoid floating state and unintentionally flipping the output.
- A quiet high and a quiet low outputs from both hex inverters to check the rail compression.
- Current limiting resistors for the LED’s
- Test points to measure the 5V, 3.3V, 555 output, Quiet High, Quiet Low, Trigger of hex inverters and test point across the resistor to measure the Voltage drop.

4. ROUGH SKETCH AND HEX INVERTER IMAGE IMAGE

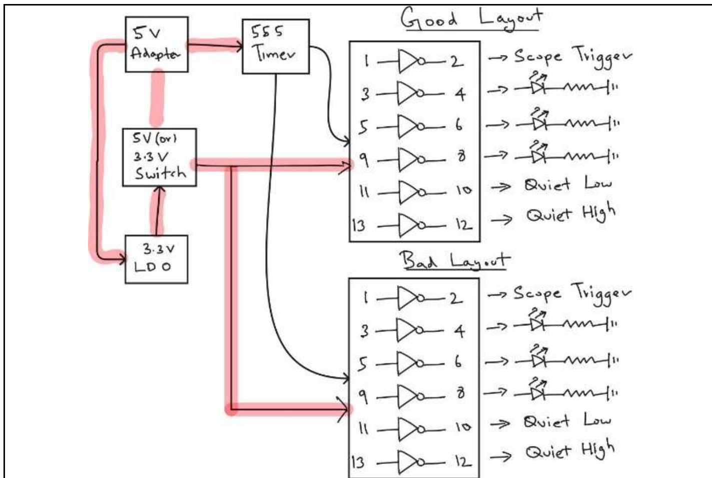


Fig 1 - Rough Sketch

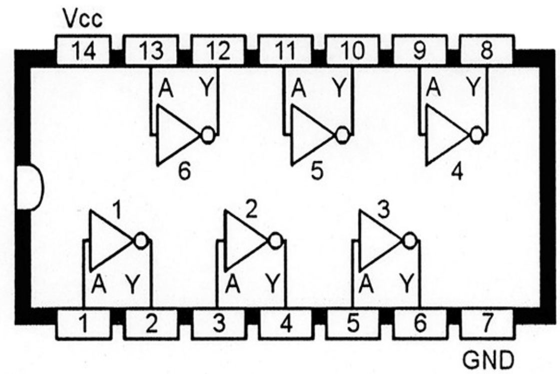


Fig 2 : Internal representation of Hex inverter

5. ESTIMATION OF COMPONENT VALUES

The 555 timer need to drive 2 Hex inverters and NE555DR sourcing and sinking capabilities are up to 200mA which satisfies our requirements. It has a output rise time = 100ns, min input supply = 4.5V

- ### FREQUENCY

Frequency (f) = $\frac{1.44}{(Ra+2Rb)*C}$ (as specified in the NE555 datasheet by Texas Instruments)

I considered **Ra & Rb = 1Kohms and C = 1uF** to get desired output frequency close to 500Hz

Substituting above values we get, $f = \frac{1.44}{(1000+2000)*10^{-6}} = 480Hz$ (which is close to our desired frequency)

- ### DUTY CYCLE

On time Duty cycle $Ton = \frac{tH}{tH+t} = 1-\left(\frac{Rb}{Ra+2Rb}\right) = 1-\left(\frac{1000}{3000}\right) = 66.6\%$

Off time Duty cycle $Toff = \frac{Rb}{Ra+2} = \frac{1000}{3000} = 33.3\%$

(as specified in the NE555 datasheet by Texas Instruments)

- ### ESTIMATION OF CURRENT THROUGH LED

Assuming the forward voltage across the LED as 2V. By Ohm's law,

Current through LED(in 5V operation) = $\frac{Voltage\ across\ current\ limiting\ resistor}{current\ limiting\ resistor\ value} = \frac{5-2}{47} = 63.82\ mA$

Current through LED(in 3.3V operation) = $\frac{Voltage\ across\ current\ limiting\ resistor}{current\ limiting\ resistor\ value} = \frac{3.3-2}{47} = 27.65\ mA$

6. Schematic Diagram

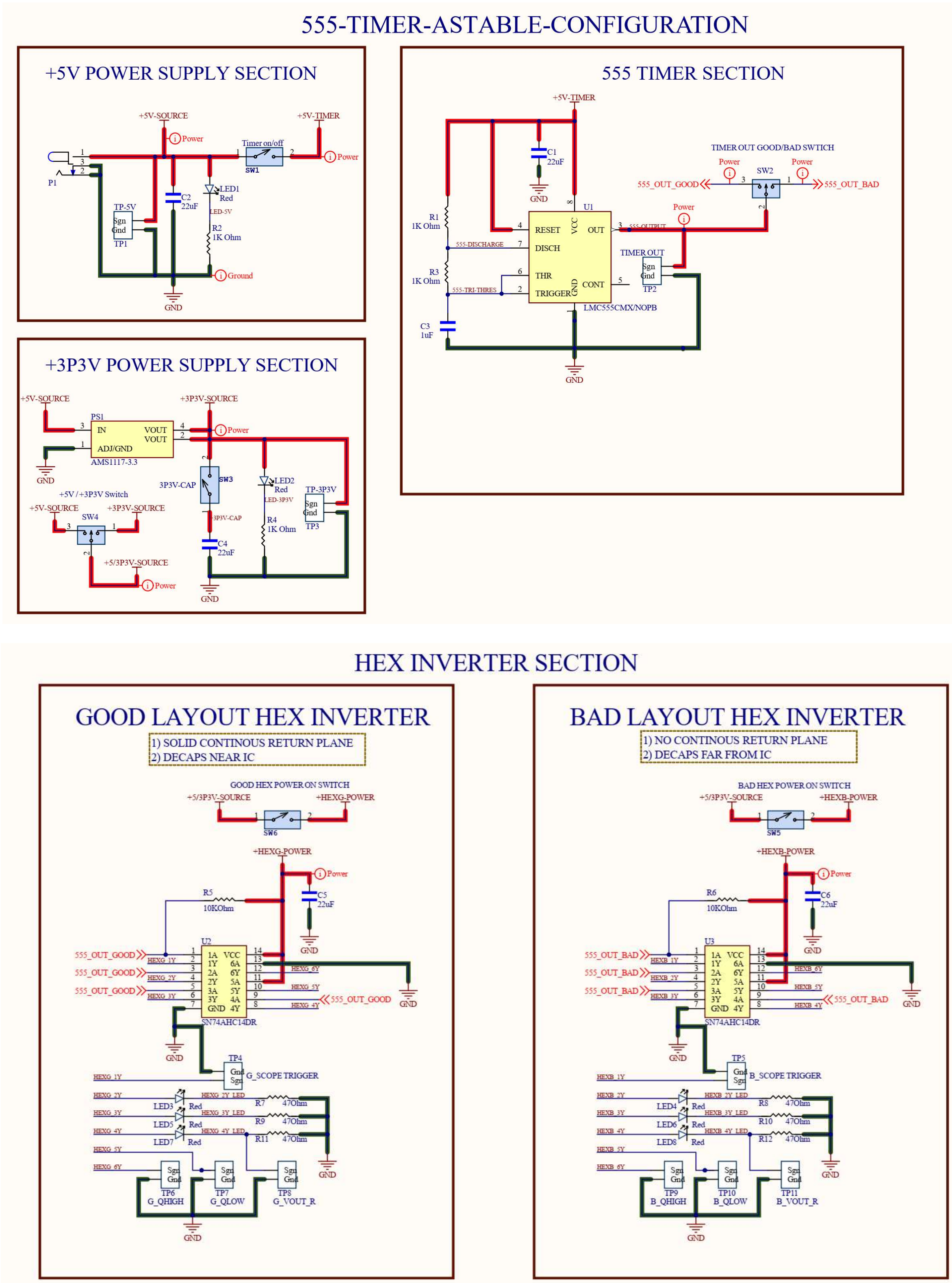


Figure 3 - Schematic diagram in Altium software

7. PCB LAYOUT

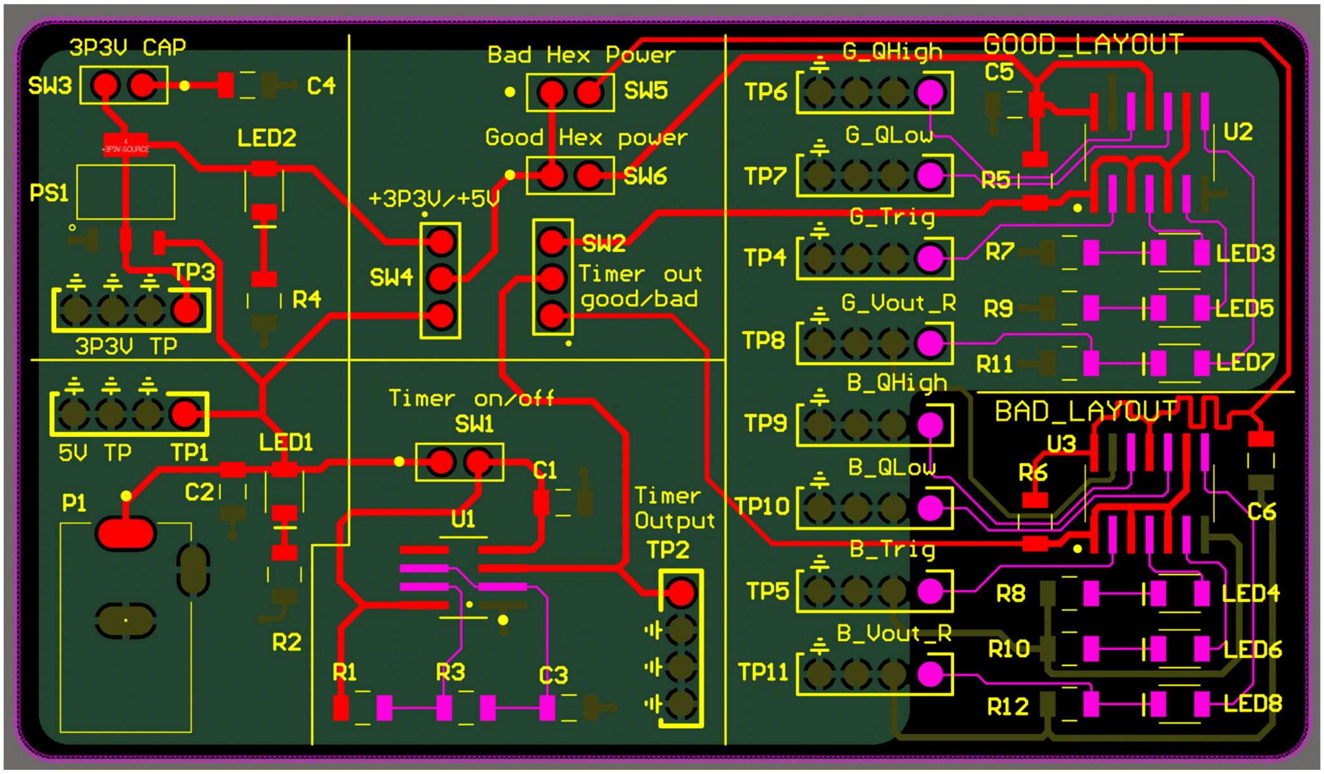


Figure 4 - PCB Layout in Altium software

Note : Except the Bad layout section(Bottom right corner) , all other sections have continuous return path and Decoupling capacitor(C6) is kept far away from bad layout hex inverter compared to good layout hex inverter.

8. BEFORE AND AFTER MANUAL ASSEMBLY

- **BEFORE MANUAL ASSEMBLY**

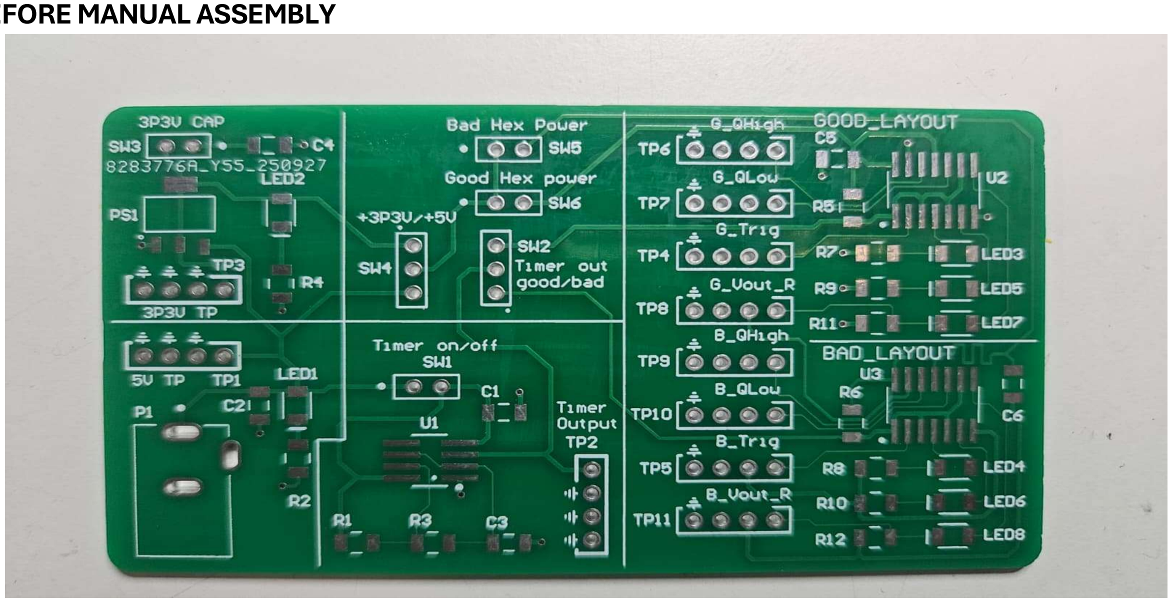


Figure 5 – Before Assembly

- **AFTER MANUAL ASSEMBLY**

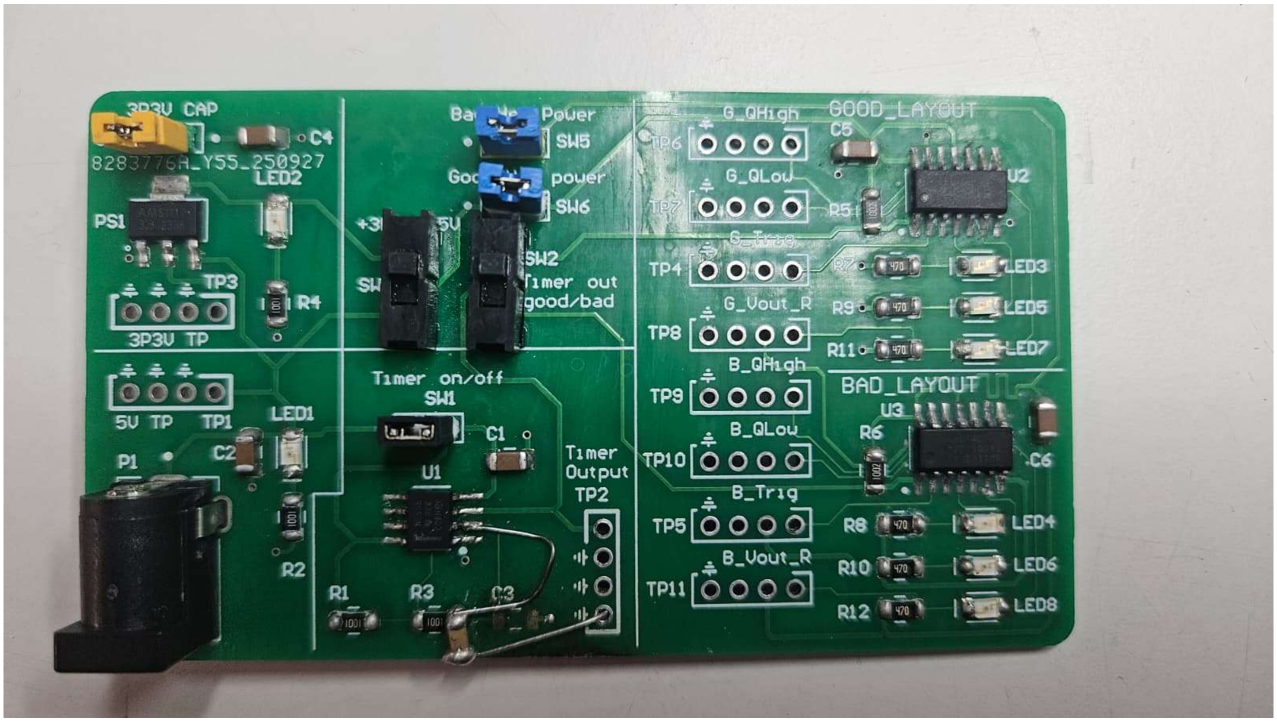


Figure 6 - After assembly

9. WORKING CRITERIA

- 5V indicator LED should turn on when powered up by 5V DC adapter and waveform should be measured in TP1
- 3.3V indicator LED should turn on and waveform should be measured in TP3
- With SW1 on, 555 timer should give a 500Hz square wave with 66% duty cycle and rise time of 100ns output measured in TP2
- SW4 should control the power rail between 5V or 3.3V.
- SW2 should control the 555 Timer output to Good or Bad layout Hex inverters.
- SW5, SW6 should control the power rail of the Good and Bad layout Hex inverters.
- Hex inverters should work and be able to light the LED in both 5V and 3.3V modes.
- Quiet Low, Quiet High and Trigger Signal from Hex inverters should be measured in their respective test points.
- Voltage through the resistor should be measured from testpoints connected to the resistors.

10. AFTER POWERING UP

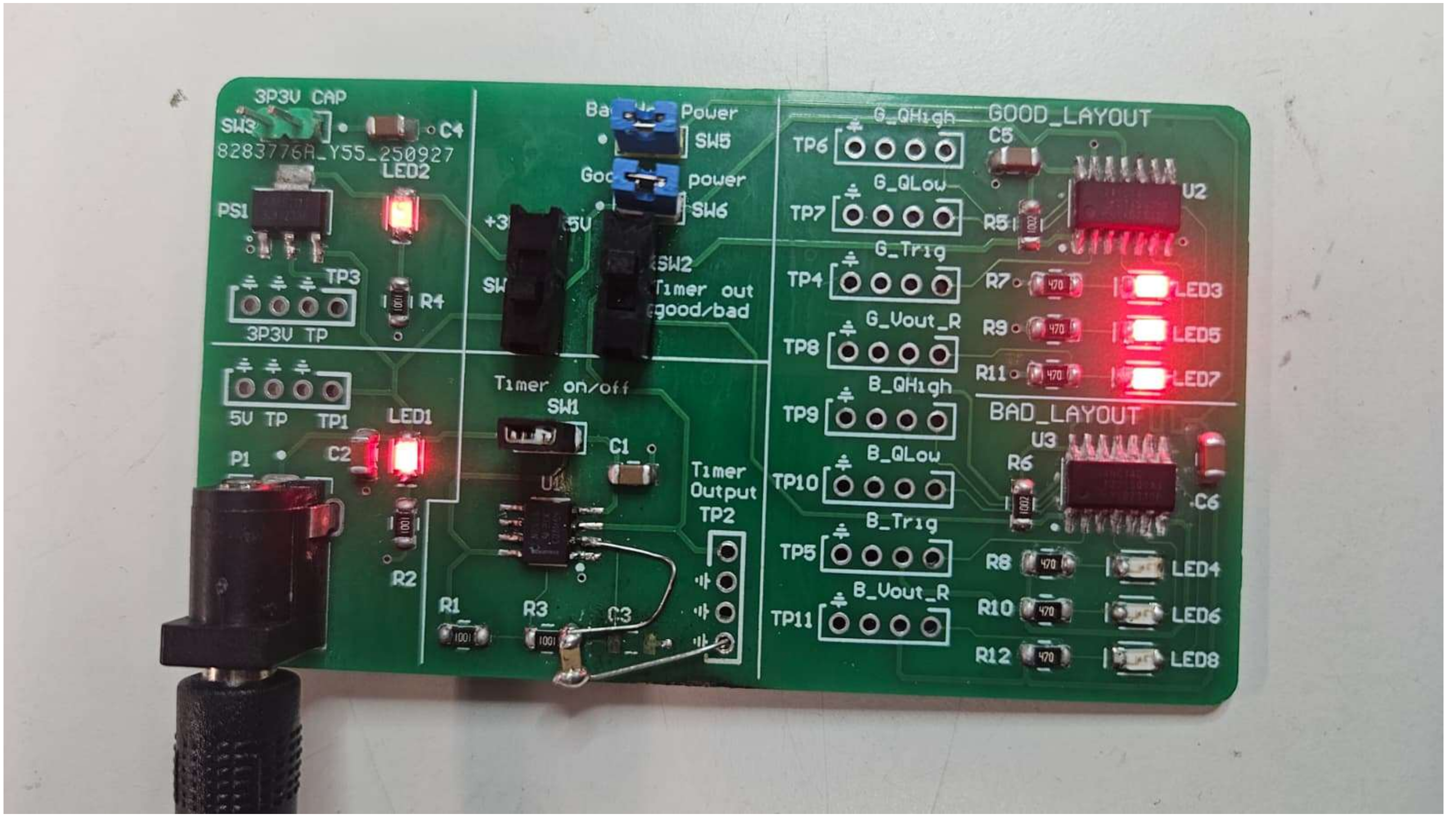


Figure 7 - Board after powering up

- The 5V indication LED lit up showing there was no issues with the power supply. Measurements were also taken using TP1 to make sure of the same.
- The 3.3V Indication LED lit up and the output was measured in the TP3
- After SW1 was turned on, output from 555 was measured using TP2 and a 5Vp-p square wave with around 500Hz and 66% duty cycle was observed.
- SW4 was used to control the voltage in the power rail and it was measured in TP4 and TP5.
- SW2 was used to control the clock input to either the good or bad layout hex inverters and the LED luminosity was observed.
- Quiet Low, Quiet High and Trigger Signal from Hex inverters was measured in their respective test points.
- Voltage through the resistor was measured from test points connected to the resistor R11 and R12.

11. DEBUGGING

The board turned on and all waveform generated had no issues and gave expected results but due to excessive heating during soldering copper pads for capacitor C3 were removed. So, jumper wires were used to solder capacitor C3 to timer U1, resistor R3 and Ground.

12. Calculating the Thevenin's Resistance

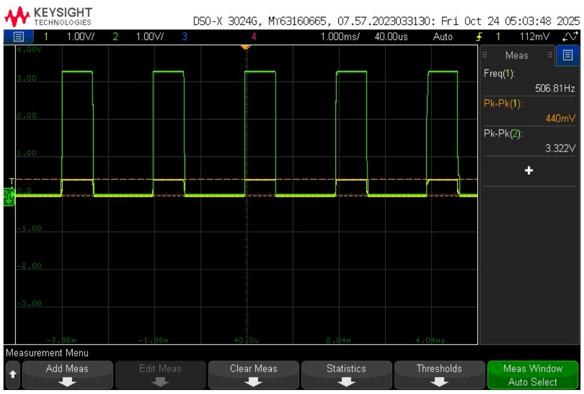


Fig 8 : Voltage across 47ohms resistor

- The amplitude of the output without load = 3.322V (open circuit voltage – V_{th})
- The amplitude of the voltage across resistor R11 = 0.440V (Resistor voltage – V_R)
- The load voltage = Diode voltage + Resistor voltage = 2V + 0.44V = 2.44V (Load voltage – V_L)
- The current through the resistor = $\frac{Resistor\ Voltage}{Resistor\ value} = \frac{0.44}{47} = 9.36\ mA$

$$Rth = \left(\frac{Vth - Vl}{IL}\right) = \left(\frac{3.322 - 2.44}{0.00936}\right) = 94.23\ ohms$$

In Figure 8, the above channel2(Green) shows the Hex inverter output measure at TP4 without Load and channel1(Yellow) shows the voltage across the 47ohms resistors when the LED is on.

This shows that the Hex inverter has an internal resistance of around **94.23 ohms** and whenever current is sourced a voltage drop occurs across this internal resistance making the Hex inverter not an ideal source.

13. 5V and 3.3V Power Rail

Fig 9 : 5v and 3.3V Power rail

Figure 9 shows 5V power rail(yellow channel) and 3.3V power rail(green channel) measured in TP1 and TP3 respectively

No issues was noticed from 3.3V output and 5V output.

14. 555 OUTPUT VS HEX INVERTERS

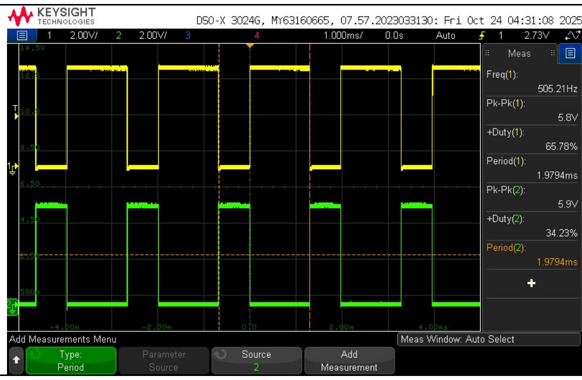


Fig 10 : 555 timer output vs Good Hex trigger signal

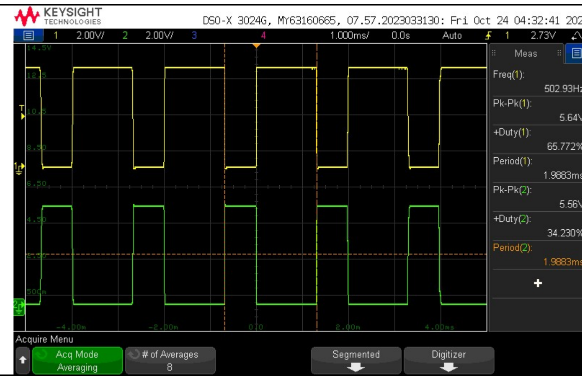


Fig 11 : 555 timer output vs Bad Hex trigger signal

Description	555 Timer out	Good Hex trigger	Bad Hex trigger
Frequency (Hz)	502	502	502
$V_{p\text{-}p}$ (V)	5.8	5.9	5.56
Duty cycle (%)	66	34.23	34.23
Period (ms)	1.98	1.979	1.983

In the above Figures, Channel 1(yellow) shows the 555 timer output and Channel 2 (green) shows the Hex inverter Trigger output. 555 clock is given as input to Hex inverter and not operation is performed. So, 555 timer output is opposite of Hex inverter trigger output. As noted from above table except duty cycle other values such as frequency, Voltage amplitude, and Period are same for both the signals.

15. EFFECT OF CAPACITOR ON 3.3V POWER RAIL

In the Below figures 12 and 13, Channel 1(yellow) shows the 555 timer output which has rise time of 79.612ns and fall time of 31 ns , Channel 2 (green) show the 3.3V power rail with 22 uF capacitor and Reference channel(Brown) shows the 3.3V power rail without capacitor during the rising and falling edges. From the data in the table we can conclude that 22uF output filter capacitor in the output of the LDO drastically reduces the voltage ripple in the power rail.

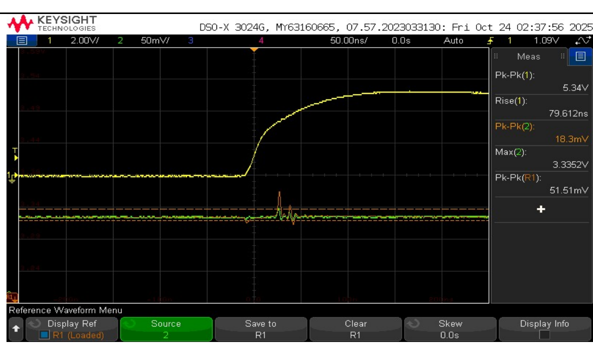


Fig 12 : Voltage ripple in 3.3V during rise time with and without 22uF capacitor

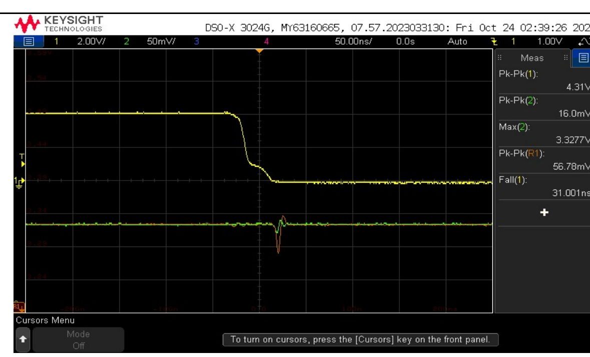


Fig 13 : Voltage ripple in 3.3V during Fall time with and without 22uF capacitor

	Voltage ripple($V_{pk\text{-}pk}$) without capacitor(brown channel)	Voltage ripple($V_{pk\text{-}pk}$) with capacitor(green channel)	Remarks
Rising edge	51.51 mV	18.3mV	Ripple has reduced by 64.47%
Falling edge	56.78 mV	16 mV	Ripple has reduced by 71.82%

16. RAIL COMPRESSION DUE TO LAYOUT DIFFERENCES – 5V

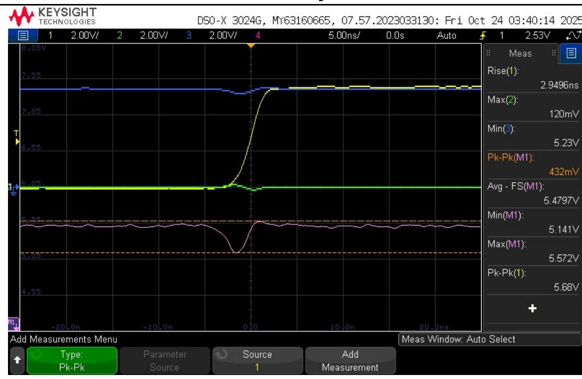
<table>
<tr><th></th><th>Good layout</th><th>Bad Layout</th></tr>
<tr><td>Rising Edge</td><td>

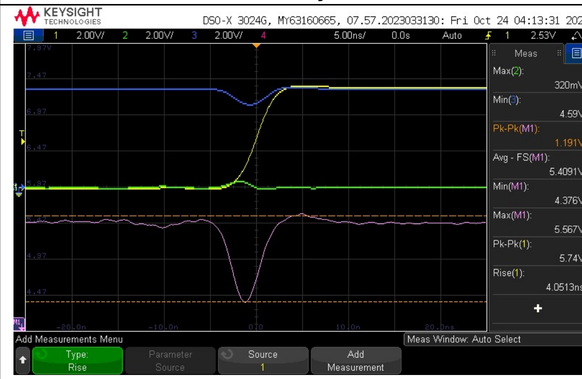
Fig 14 : Rail compression during rise time in good layout</td><td>

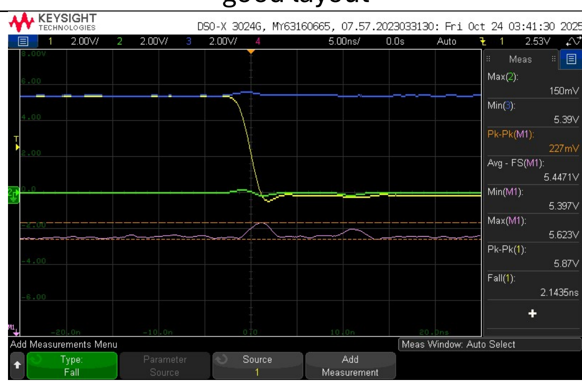
Fig 15 : Rail compression during rise time in Bad layout</td></tr>
<tr><td>Falling edge</td><td>

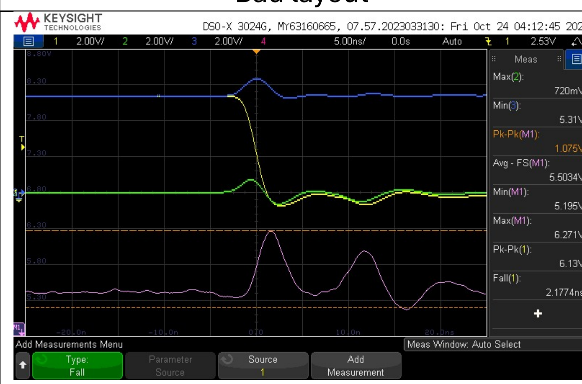
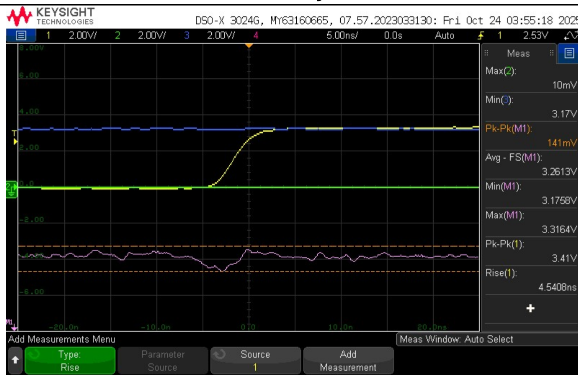
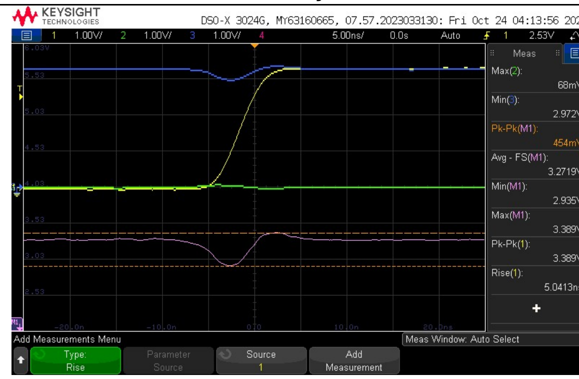
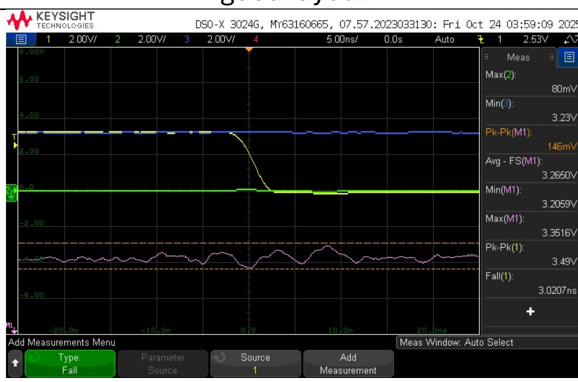
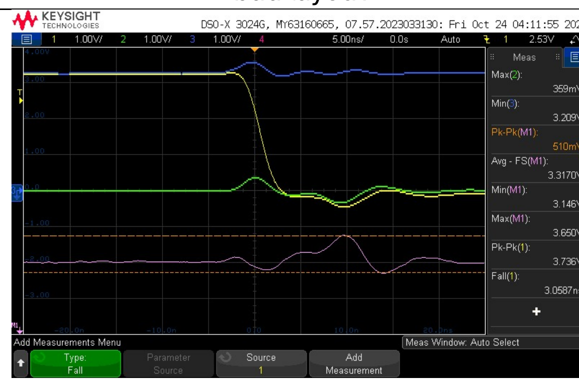
Fig 16 : Rail compression during fall time in good layout</td><td>

Fig 17 : Rail compression during fall time in bad layout</td></tr>
</table>

	Good Layout	Bad layout	Remarks
Rise time	2.94 ns	4.051 ns	Bad layout rise time is slower compared to Good layout
Fall time	2.14 ns	2.17 ns	Both fall times are similar
Rail compression during rise time(V_{pk-pk})	432 mV	1.191 V	175.69% increase in rail compression.
Rail compression during Fall time(V_{pk-pk})	227 mV	1.075 V	373.57% increase in rail compression.

In the figures 14,15,16 & 17, **yellow** is the hex-inverter trigger, **blue** is Quiet High (QH), **green** is Quiet Low (QL), and **pink** is the **rail-compression** trace defined as QH − QL—the effective supply seen by the inverter (VCC–GND). In the **non-transient** region, rail compression is minimal and the pink trace stays near the nominal VCC. During **transitions**, voltage droop (on VCC) and ground bounce (on GND) compress the rail, reducing the true inverter supply. Measured extremes of the pink trace are: **good layout** → minimum **5.141 V**, maximum **5.623 V**; **bad layout** → minimum **4.376 V**, maximum **6.271 V**.

17. RAIL COMPRESSION DUE TO LAYOUT DIFFERENCES – 3.3V

	Good layout	Bad Layout
Rising Edge	 Fig 18 : Rail compression during rise time in good layout	 Fig 19 : Rail compression during rise time in bad layout
Falling edge	 Fig 20 : Rail compression during fall time in good layout	 Fig 21 : Rail compression during fall time in bad layout

	Good Layout	Bad layout	Remarks
Rise time	4.54 ns	5.041 ns	Bad layout rise time is slower compared to Good layout
Fall time	3.02 ns	3.05 ns	Both fall times are similar
Rail compression during rise time(V_{pk-pk})	141 mV	454 mV	221.99% increase in rail compression.
Rail compression during Fall time(V_{pk-pk})	146 mV	510 mV	249.32% increase in rail compression.

In the figures, **yellow** is the hex-inverter trigger, **blue** is Quiet High (QH), **green** is Quiet Low (QL), and **pink** is the **rail-compression** trace defined as QH − QL—the effective supply seen by the inverter (VCC–GND). Like the 5V operation, In the **non-transient** region, rail compression is minimal and the pink trace stays near the nominal VCC. During **transitions**, voltage droop (on VCC) and ground bounce (on GND) compress the rail, reducing the true inverter supply. Measured extremes of the pink trace are: **good layout** → minimum **3.17 V**, maximum **3.35 V**; **bad layout** → minimum **2.935 V**, maximum **3.65V**. The increase in rise time compared to 5V operation is due to slower charging time of the overall capacitance in the system.

18. Best design practices followed

1. Decoupling capacitor is placed near the IC to minimize voltage noise due to the voltage drop across the loop inductance in the power rail except for the bad layout to observe the difference between them.
2. Proper labelling was given to all switches and test points for better understanding of the board
3. Switches were placed between different sections and proper indicating LEDs were placed to aid for debugging
4. Continuous ground plane was provided on bottom layer for proper return path except for bad layout and thicker trace widths were used for power nets.
5. No of unique parts was reduced.

19. Improvements to be implemented

Have to be more cautious during manual soldering so that PCB is not damaged.

20. Conclusions

This board study practically validates key design practices—placing decoupling capacitors close to the IC and maintaining a continuous return plane beneath the signal—and it also enabled direct measurement of on-die power-rail noise. The internal Thevenin's resistance was also calculated which was around 94.23 ohms.